

PROJECT-BASED LEARNING

PHASE-I EVALUATION

Project Title

Team ID: XXXX

Department of Computer Science & Engineering

Graphic Era (Deemed to be University), Dehradun

Academic Session 2026-27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T043
Semester: 3rd
Section: XX
Domain: cleanlink

Team Members

1. *Raj Royal – 2510370233*
2. *Aditi Khanka – 2510380023*
3. *Lopamudra Batabyal – 2510370101*
4. *Vedant Nautiyal – 2510011197*

Mentor

Mr. Priyanshu Rawat

Interactions completed: _____

Problem Statement & Motivation

Problem Statement

Waste collection often relies on fixed schedules and manual monitoring, leading to delayed cleaning, overflowing waste spots, and inefficient collection trips.

Motivation

- ▶ *Current Gap: Unorganized waste reporting.*
- ▶ *Impact: Delayed cleaning and job allocation.*
- ▶ *Limitation: Fixed schedules and manual monitoring*

Target Users / Stakeholders

- ▶ *Primary Users: Citizens, Workers, Funders*
- ▶ *Secondary Stakeholders: Verification Team, Admin .*
- ▶ *Expected Benefits: Faster cleaning, organized jobs, community participation*

Objectives

Project Objectives

1. *Develop a platform for reporting and managing waste spots.*
2. *Implement physical verification of reported waste.*
3. *Connect verified waste with community funding.*
4. *Create and assign waste-collection micro-jobs.*
5. *Integrate DSA and OOP concepts to build and evaluate the system.*

Key Objective: *To convert verified waste reports into community-funded micro-jobs and efficiently assign them to suitable workers.*

Proposed Solution

Proposed Approach

1. *Input: Photo, location and estimated waste size*
2. *Processing / Core Algorithm: Verify waste size prioritize jobs*
3. *System Logic: Report → Verify → Fund → Assign → Complete*
4. *Database / Storage: Store users, waste spots jobs*
5. *Output / Visualization: Verified waste details job status*

Key Features

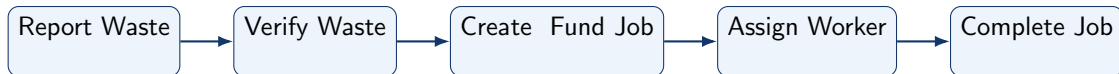
- ▶ *Waste reporting*
- ▶ *Physical waste verification*
- ▶ *Community funding*
- ▶ *Micro-job creation*
- ▶ *Worker assignment*

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	<i>Linked Lists, Queues, Priority Queues, Hash Tables, Arrays + Sorting, Graphs (Haversine formula), File Handling</i>	<i>Data Engine – Reporting, Job & Funding, Worker Assignment</i>
OOPs with C++	<i>Classes/Objects, Inheritance, Polymorphism, Encapsulation, Composition, Abstraction</i>	<i>Application Layer – User, WasteSpot, Job, Campaign</i>

Requirement: Demonstrate meaningful integration of both subjects in **one** project.

System Architecture / Workflow



Major Components

- ▶ *Waste Reporting Module*
- ▶ *Verification Module*
- ▶ *Job Funding Module*
- ▶ *Worker Assignment Module*
- ▶ *Database Admin Module*

Data / Control Flow

- ▶ *Waste data moves from reporting to verification*
- ▶ *Verified data moves to job creation and funding ,Funded jobs are assigned to workers*
- ▶ *Output: Completed job with updated status*

Technology Stack & Methodology

Technology Stack

- ▶ Programming: C (data engine) / C++ (application layer)
- ▶ Storage: File handling – .dat files (no DBMS used)
- ▶ Interface: Command-Line Interface; ncurses/Qt if time permits
- ▶ Version Control: Git / GitHub

Development Methodology

1. C data structures & file persistence
2. C++ class hierarchy (built on the C layer)
3. End-to-end workflow: Report → Verify → Fund → Assign → Complete
4. Funding, reputation (CleanScore) & dashboard
5. Testing, edge cases & integration

Bottom-up build order: each layer is tested independently before integration.

Initial Progress & Team Contribution

Progress Achieved

- ▶ *Background / literature study*
- ▶ *Requirements identified*
- ▶ *Architecture prepared*
- ▶ *Initial module implemented*
- ▶ *Database / data preparation*

Individual Contribution

Member	Role	Contribution
M1	Lead	XXXX
M2	Developer	XXXX
M3	DB/Testing	XXXX
M4	Developer/Docs	XXXX

Roadmap, Expected Outcomes & References

Roadmap

1. Core data & class layer (C/C++)
2. Core workflow: Report → Verify → Fund → Assign → Complete
3. Funding, reputation & dashboard
4. Testing, polish & demo prep

Expected Outcomes

- ▶ Working console prototype of full cleanup cycle
- ▶ Real DSA + OOP applied at every stage
- ▶ CleanScore reputation system & dashboard
- ▶ Practical community waste-management model

References

1. Swachh Bharat Mission (MoHUA)
2. CPCB – Municipal Solid Waste Reports
3. Ketto & Milaap – Crowdfunding models
4. Urban Company – Gig-work assignment model

Thank You
Questions & Suggestions